Governing Artificial Intelligence in the European Union: Risk, Ethics, and the Architecture of the AI Act

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Abstract

The Artificial Intelligence Act (EU Regulation 2024/1689) is the first comprehensive attempt to regulate AI at the European level through a risk-based approach grounded in fundamental rights. This paper examines how the Act, with its balanced approach, strikes a reassuring equilibrium between normative ambition and practical enforceability. It delves into three key aspects: the political and legal framing of the regulation, the classification and scope of prohibited practices, and the role of ethical principles such as "Ethics by Design." Through this analysis, the paper highlights unresolved tensions between ethics, legal effectiveness, and technological competitiveness, questioning whether the Act can deliver more than symbolic governance.

CCS Concepts

• **AI regulation and legal frameworks; Ethical and political implications of AI decision-making**;

Keywords

Law, AI-Act, Artificial Intelligence, Risk, Ethics

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1 Introduction

The Artificial Intelligence Act (Regulation EU 2024/1689) marks a significant milestone as the first comprehensive attempt to regulate artificial intelligence within the European Union and beyond. Its pioneering risk-based approach, underpinned by the protection of fundamental rights, is a crucial strategy in promoting the ethical, safe, and sustainable development of emerging technologies. This paper aims to dissect three core dimensions of the AI Act: its legal and political framework, the definition and scope of prohibited practices, and the normative role of ethics, particularly through the concept of "Ethics by Design." The aim is to highlight the internal tensions within the regulation, between ambition and applicability, and to assess the coherence of the European model in striking a balance between technological innovation and the protection of fundamental rights. While the paper is structured around the main thematic pillars of the AI Act, its purpose is not to offer a neutral or merely descriptive overview. Instead, each section adopts a critical perspective: the goal is to assess not only the declared intent of the regulation but also its conceptual vagueness, operational limitations, and the unresolved tensions between ethical principles and legal enforceability.

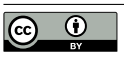
2 Legal framework and scope of the AI-Act: an 'ambitious' objective

The legal and political genesis of the AI Act is a clear demonstration of the EU's leadership claim in shaping global standards in AI governance. However, this claim is not without its challenges, particularly in the face of the rapid evolution of technology.

The European choice has not prioritised the competitiveness of artificial intelligence, which is inevitably compromised due to the unbridgeable technological gap with countries such as the United States and China, but rather its ethical and sustainable promotion [3]. The goal is not only to regulate the use of AI, but also to prevent and punish abuses. This urgency was already evident in 2017, when the European Council proposed a dedicated regulation [4]. A similar vision was also adopted by the European Parliament, which developed a significant strategy that began with soft-law initiatives, among which the joint creation of the so-called "European Strategy on Artificial Intelligence" stands out, later culminating in the publication of the "White Paper". After consultations with stakeholders, the Parliament and Council quickly reached a final compromise on the draft regulation. Although this compromise was achieved swiftly, the legislative debate centered on "foundation models"[1]. At the same time, the European Parliament has proposed requiring that providers of AI systems classified as "high risk" proactively conduct impact assessments on fundamental rights. This measure aims to verify the level of risk and necessitate the adoption of tailored and appropriate mitigation strategies.

Notably, the AI ACT is part of a broader and systemic legislative framework and is encompassed within the "New Legislative Framework", which was established to ensure adequate product compliance. For instance, the AI Act shares specific provisions with the GDPR, such as the concept of risk. Therefore, the issue of risk, concerning algorithmic reliability, is crucial, as it determines its evolution [11]. Indeed, the prevailing doctrine has extensively argued that data protection is a key point, leading to the broadest "algor-ethics" [8].

[1] It was precisely the protection of fundamental rights that was the central issue, the crossroads, in the bargaining between members of Parliament, the Commission and the Council



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Nonetheless, while the Act seeks to introduce a binding legal framework, several ambiguities remain unresolved. These tensions undermine the clarity and legal certainty that such a pioneering regulation should provide. Firstly, the regulation presents critical issues related to the risk of "obsolescence", despite the periodic evaluations[2] that are to be carried out by the Commission [6][3]. Secondly, the AI Act does not provide a clear definition of artificial intelligence. Instead, consider an AI system "an automated system designed to operate with varying levels [6]." In this sense, Recital 12 clarifies that "techniques that enable inference in the construction of an AI system include machine-learning approaches that learn from (i) data; (ii) how to achieve certain goals; (iii) logic- and knowledge-based approaches that draw inferences from codified knowledge." Consequently, the significant development is the implementation of machine self-learning. This broad and open-ended definition reflects the EU's attempt to future-proof the regulation but also generates uncertainty for developers and enforcers. Thirdly, Art. 2 of the AI Act clarifies that the regulation applies "to suppliers that place on the market or commission AI or place AI models for general purposes in the Union, regardless of whether they are established or located in the Union or a third country." Therefore, the regulation applies not only to deployers of AI systems established in the EU [6] or present in its territory, but also to suppliers and/or deployers based in a third country, provided that the output is received in the EU [6][4]. Based on the principle of risk, European legislators have imposed stricter limitations on AI due to potential human rights violations. Nevertheless, the actual implementation of such cross-border obligations poses serious questions of jurisdictional reach, legal enforceability, and regulatory cooperation with non-EU actors.

These dimensions remain largely unaddressed in the current version of the Act. These foundational choices reveal a complex tension at the heart of the EU's AI governance strategy. On the one hand, the EU is striving to establish itself as a normative leader in AI governance. On the other hand, it must navigate the constraints imposed by its technological competitiveness and the existing legal framework.

3 Risk gradability and AI practices deemed unacceptable

The prohibited practices in the AI Act represent the EU's commitment to protecting fundamental rights by translating ethical principles into enforceable legal rules. This commitment, however, is not without its challenges, as it must navigate through definitional vagueness and enforcement uncertainty.

Article 5(a) of the AI Act centers, which contains the first prohibited practice, is focused on the notion of significant harm, prohibiting the placing on the market, putting into service, or use of AI systems that "employ subliminal or deliberately manipulative techniques" capable of distorting a person's or group's behaviour substantially. This vagueness is expected to spark substantial legal and academic debate, especially concerning harmonized enforcement. Nonetheless, by Recital 29 of the AI Act, the European legislator has defined significant harm only when there are "sufficiently significant" adverse effects. This should permit a detailed rule at the national level [12][5]. Yet, such decentralization risks producing interpretive fragmentation among Member States, undermining regulatory consistency.

There is also a debate for the third prohibited practice, which is indicated in (c) of the same article 5, which introduces the prohibition of social scoring. Such practices include classifying individuals or groups based on social behaviour or personal characteristics, whether actual or expected. This standard highlights the significant cultural and legal gap between Europe and other parts of the world, such as China, where similar practices are still permitted. Authors such as R.J. Neuwirth have pointed out that the absence of such a ban could have led to the implementation of mass control, as already observed in the Chinese context. Nonetheless, emphasising these principles, the European legislator, in Council Regulation No. 31, defines the classification practices as violating the right to dignity, discrimination, equality, and justice. These social scoring practices are also extended to private individuals, stating that the prohibition applies as the social score is detrimental. The scope of this prohibition remains controversial, especially in contexts where reputational scoring systems already exist (e.g., credit scores), raising questions on the distinction between social control and risk-based classification.

The fourth prohibited by Article 5(d) of the AI Act is real-time biometric recognition. The European legislator initially provided biometric recognition, accompanied by special safeguard clauses. However, in the final version, the co-legislators adopted a more restrictive approach, introducing its prohibition, albeit with some mandatory exceptions. These include the use of biometric recognition for the targeted search of potential crime victims, such as in cases of kidnapping, sexual exploitation of human beings, or missing persons. The permissible use is limited to situations involving a specific, substantial, and imminent threat to the safety or life of individuals, such as actual or foreseeable terrorist attacks or the execution of a European Arrest Warrant. This may also include suspects if there is a well-founded basis for doing so. [6][6]. The application of these cases is subject to authorisation by the competent judicial authority or an independent administrative authority of the Member State. Requirements of necessity and proportionality must justify authorisations, except in cases of urgency, in which granting authorisation is deferred with retroactive effect [9][7]. Despite the apparent stringency of the prohibition, real-time biometric recognition is one area where the gap between the AI Act's normative ambition and its actual enforceability becomes most evident. While the regulation includes only narrowly defined exceptions, these may expand in practice, particularly in emergency contexts. However, it is the transnational dimension that exposes the most critical vulnerabilities. In a digital ecosystem dominated by global actors and distributed cloud infrastructures, the effectiveness of the ban is undermined by the lack of binding cross-border enforcement mechanisms.

[2] From 2028, an update every four years
[3] Art. 112 of the AI-Act.
[4] Art. 2(1)(c) of the *AI Act.*
[5] DDL-n.1146accessible, at: https://www.senato.it/leg/19/BGT/Schede/
[6] Art. 5(2) of the *AI Act.*
[7] M. IASELLI, *op.* cit.., p. 64.

The fragmentation of national supervisory authorities, the diversity of sanctioning regimes, and the absence of centralized oversight at the EU level make it extremely difficult to counteract prohibited practices that originate or operate outside a single Member State's jurisdiction. The risk is that facial recognition systems are banned in

4 Ethics and AI - Act principles

The ethical framework of the AI Act, particularly the emphasis on 'Ethics by Design', is key to understanding the EU's normative ambition. Nevertheless, ethics, once institutionalised, may risk becoming decorative unless backed by enforceable and coherent mechanisms. Ethics applied to artificial intelligence aims to address the moral issues associated with information, data, and algorithms, developing solutions that promote good conduct [1].

In this context, ethics is divided into complex ethics and soft ethics [7]. The former is represented by what we usually have in mind when we discuss values, rights, duties, and responsibilities; the latter is understood as what we typically have in mind regarding post-compliance ethics, providing a strategic opportunity that enables actors to exploit the social value of digital technologies and manage risk [2]. Nonetheless, this dual structure of ethics also raises concerns: while complex ethics defines clear normative obligations, soft ethics may operate more as a reputational tool than as a constraint, allowing broad discretion in its actual implementation. The doctrinal debate has long pointed out that artificial intelligence systems are not "intelligent". For this reason, ethical issues related to fairness, data protection, accountability, bias, and algorithmic transparency arise. Therefore, the legislative framework of the AI Act must address ethical concepts and principles centred on the safe, reliable, and sustainable development of artificial intelligence [3].

Since 2018, EU institutions have repeatedly reaffirmed the need for a coherent, ethical, and legal framework for AI, one that aligns with European values. In any case, today, it is possible to speak of ethics "made in Europe", understood as ethics from design – "Ethics by design" - to promote both ethical and innovative aspects, thereby providing a competitive advantage to companies operating from and for the European market [9][8]. The regulation is inspired by the principles of "Ethics by design". It aims to embed respect for the rule of law, fundamental rights, and diversity into AI systems, while promoting a competitive European AI ecosystem [2]. Nevertheless, despite these references, the AI Act does not specify how ethical compliance should be measured or enforced, leaving ethical commitments vulnerable to remain purely aspirational.

Another proposed ethical principle in this perspective is "allowing all individuals an equal and fair chance to access the available opportunities of a society." This signifies that individuals or groups in comparable situations should not be treated less favourably solely due to a "specific characteristic, such as gender, racial or ethnic origin, belief, disability, age, or sexual orientation".

Ultimately, the Act's ethical architecture reflects the EU's ambition to lead by its values. Regardless, the lack of operational mechanisms, such as binding indicators, oversight bodies, or sanctions, raises doubts about the enforceability and consistency of these ethical principles. Once the ethical criteria governing artificial intelligence systems have been analysed, it is helpful for this paper to return to our starting point, namely, to evaluate the role of sustainability in artificial intelligence systems. This exposes a key contradiction: ethics, if not supported by enforceable structures, may serve as little more than a symbolic gesture rather than a foundation for real accountability.

[8] M. IASELLI, *op.* cit., p. 113.

5 Concluding Remarks

The Artificial Intelligence Act, a pioneering endeavour by the European Union, sets out to regulate artificial intelligence through a legal framework rooted in fundamental rights, transparency, and sustainability. This regulation not only aims to exercise a form of moral and democratic leadership at the global level but also proposes an alternative model to that of technologically dominant powers, albeit less bound by the principles of the rule of law. As discussed in the first section, this regulatory ambition is based on a vision consistent with the Union's values. However, it grapples with a series of structural ambiguities, from the definition of artificial intelligence to the actual applicability of the concept of risk. The tension between declared ethics and regulatory feasibility, a hallmark of European design, underscores the complexity of this regulatory endeavour. The second section, dedicated to practices with 'unacceptable risk,' underscores the interpretative limitations of central notions such as 'significant harm' or 'vulnerability.' While these notions are crucial for the protection of rights, their legal vagueness leaves broad discretion to Member States. This highlights the challenges the AI Act faces in ensuring uniform application at the European level. The third section demonstrated how the ethical dimension of the AI Act, while formally central, remains largely entrusted to declarative instruments and general principles, without binding mechanisms or concrete evaluation indicators. the AI Act provides a crucial foundation for responsible technological development aligned with European values. However, its success will depend on the ability to transform value statements into effective, uniform, and measurable legal instruments. The question remains whether the regulation will truly succeed in establishing itself as a global model or, conversely, whether it will reveal the intrinsic limitations of attempting to legislate innovation and rights in a rapidly changing context.

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